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UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**SEKOLAH
KOMPUTERAN**

SEMESTER I 2019/2020

COURSE CODE : SCSD2613

COURSE NAME : SYSTEM ANALYSIS AND DESIGN

YEAR / PROGRAM : 1 / 2 (SCSB/SCSJ/SCSR/SCSV)

TIME :

DATE :

VENUE :

INSTRUCTIONS TO CANDIDATES:

1. Fill in your particulars in the question booklet and answering booklet:

Name	AKMAL RAFFAEL BN AHMAD RAFFAEL
Passport No.	
Matric No.	A2550181
Year / Program	1 / 2 SCSPH
Lecturer	

2. Read the instructions given for each question carefully.
3. Answer **ALL** questions in this question paper.
4. This paper consists of two parts;
Part A: 4 STRUCTURED questions.
Part B: 1 CASE STUDY question.
-

THIS PAPER CONSISTS OF EIGHT (8) PRINTED PAGES, INCLUDING THIS PAGE.

PART A : STRUCTURED QUESTIONS (70 MARKS)

INSTRUCTION : ANSWER ALL QUESTIONS.

Please write your answer in the answer sheets.

QUESTION 1 (20 marks)

- a) A marketing company wishes to construct a decision table to decide how to treat clients according to three characteristics: Gender, City Residents, and age group: (A (under 30), B (between 30 and 60), C (over 60)). The company has four products (W, X, Y and Z) to test market. (Product W will appeal to female city residents. Product X will appeal to young (under 30) females. Product Y will appeal to Male middle-aged (group B) shoppers who do not live in cities. Product Z will appeal to all but older (group C) females.)

- i. Construct a decision table based on (a).

(6 marks)

- ii. Simplify the decision table in (i).

(4 marks)

- iii. Do you think decision table is more helpful to analyse the decision to treat the clients in Question 1, compared to decision tree? Why?

(2 marks)

- b) Your company is deciding on whether to develop and launch a new product. Research and development costs are high and there is a chance that the product launch will be successful, or it will fail. The level of expected profit is as follows:

- if the level of expected profit is RM800,000 per year, then it is considered as HIGH investment,
- if the level of expected profit is RM500,000 per year, then it is considered as MEDIUM investment, and
- if the level of expected profit is RM200,000 per year, then it is considered as LOW investment

Any expected profit that is less than RM200,000 per year is considered as a failure and not worth to make investment at all.

Construct a **decision tree** to show the investors how to decide the investment level.

(8 marks)

A marketing company wishes to construct a decision table to decide how to treat clients according to three characteristics: Gender, City Residents, and age group: (A (under 30)) (B (between 30 and 60)) (C (over 60)). The company has four products (W, X, Y and Z) to test market. (Product W will appeal to female city residents.) Product X will appeal to young (under 30) females. Product Y will appeal to Male middle-aged (group B) shoppers who do not live in cities. Product Z will appeal to all but older (group C) females.)

Condition	Rules											
	1	2	3	4	5	6	7	8	9	10	11	12
gender	F	F	F	F	F	F	M	M	M	M	M	M
city resident	Y	Y	Y	N	N	N	Y	Y	Y	N	N	N
age group	A	B	C	A	B	C	A	B	C	A	B	C
Action												
Product W	X	X	X									
Product X	X			X								
Product Y										X		
Product Z	X	X		X	X		X	X	X	X	X	X

Condition	Rules											
	1	2	3	4	5	6	7	8	9	10	11	12
gender												
city resident												
age group												
Action												
Product W												
Product X												
Product Y												
Product Z												

gender → 2 → M/F
city resident → 2 → Yes/no
age group → 3 → A/B/C

2 x 2 x 3 → 12 rules

Conditions and Actions				Rules							
				1	2	3	4	5	6	7	8
Customer ordered from Fall catalog.				Y	Y	Y	Y	N	N	N	N
Customer ordered from Christmas catalog.				Y	Y	N	N	Y	Y	N	N
Customer ordered from specialty catalog.				Y	N	Y	N	Y	N	Y	N
Send out this year's Christmas catalog.				X	X	X	X	X	X	X	X
Send out specialty catalog.											
Send out both catalogs.				X				X			
Conditions and Actions				Rules							
				1'	2'	3'	4'	5'	6'	7'	8'
Customer ordered from Fall catalog.				Y	Y	Y	Y	N	N	N	N
Customer ordered from Christmas catalog.				Y	Y	N	N	Y	Y	N	N
Customer ordered from specialty catalog.				Y	N	Y	N	Y	N	Y	N
Send out this year's Christmas catalog.				X	X	X	X	X	X	X	X
Send out specialty catalog.											
Send out both catalogs.				X				X			

M M M M
N Y - -
B - A C
X X X X

F F F F F
Y Y Y Y Y
A C A A A
X X X X X
, A B
X

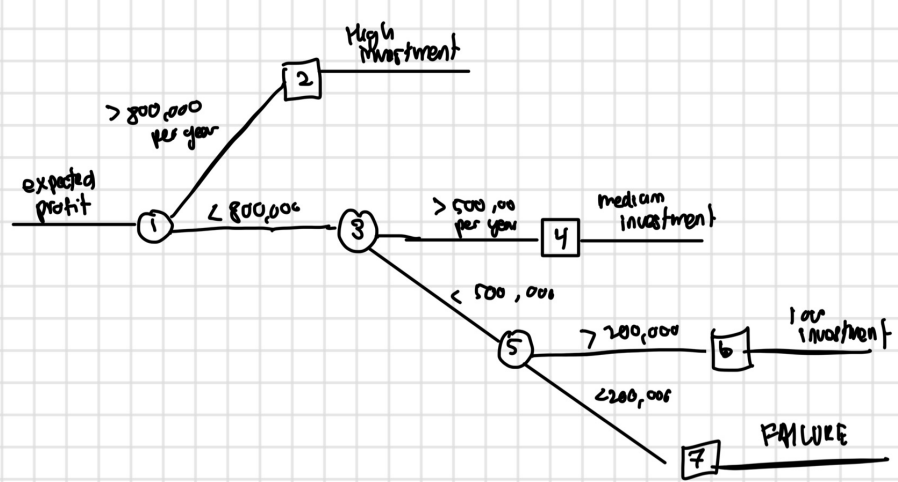
iii) yes, because in this situation, if have really complex combination of condition, using a decision table will visualize it better. While decision tree is more suited to sequence of action or condition.

b) Your company is deciding on whether to develop and launch a new product. Research and development costs are high and there is a chance that the product launch will be successful, or it will fail. The level of expected profit is as follows:

- if the level of expected profit is RM800,000 per year, then it is considered as HIGH investment,
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QUESTION 2 (15 marks)

- a) Explain **TWO (2) differences** between Data Flow Diagram and Structure Chart?
(3 marks)
- b) Figure 1 (a-c) below shows the logical data flow diagram of Food Ordering System.
Convert the logical data flow diagram below into a structure chart.
(12 marks)

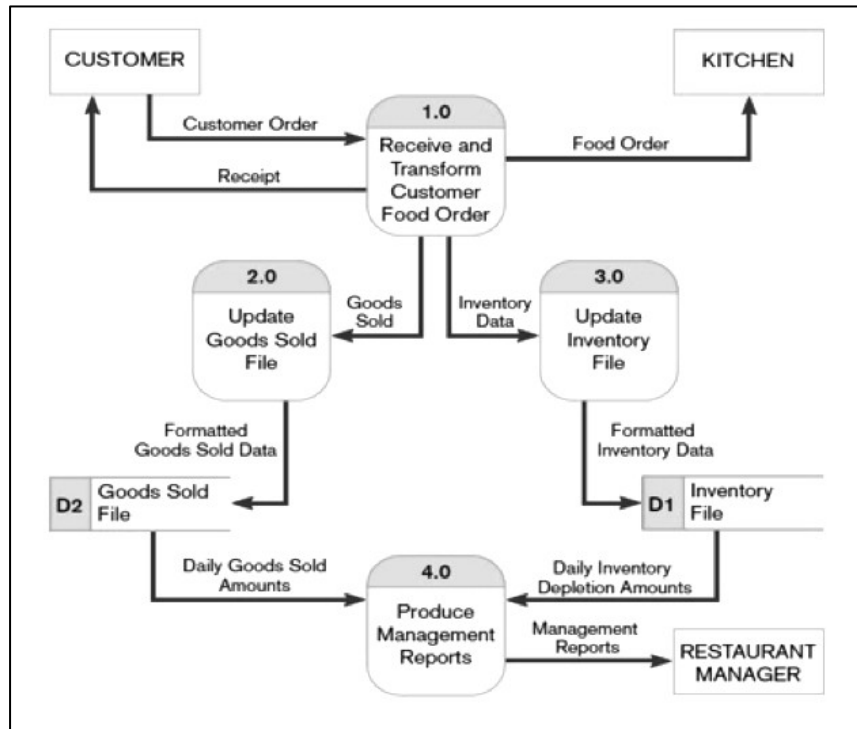


Figure 1(a): Level 0 Diagram for Food Ordering System

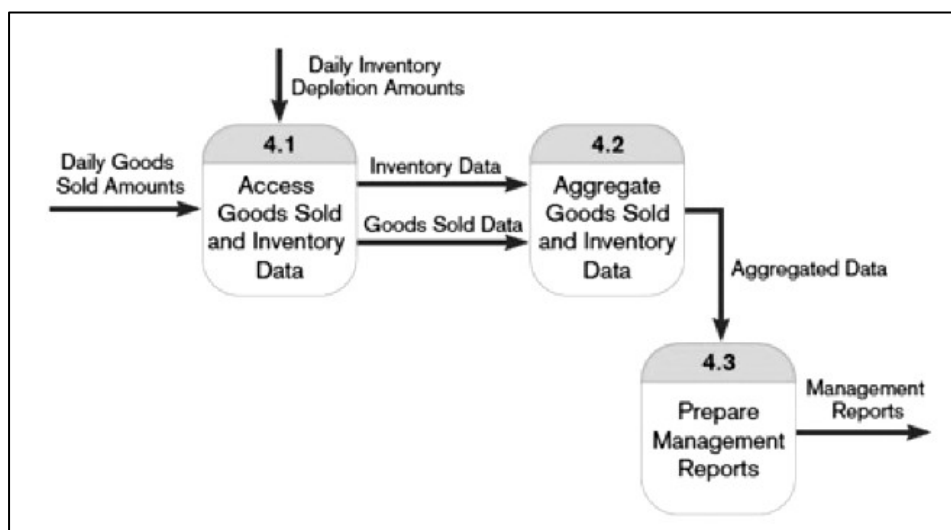
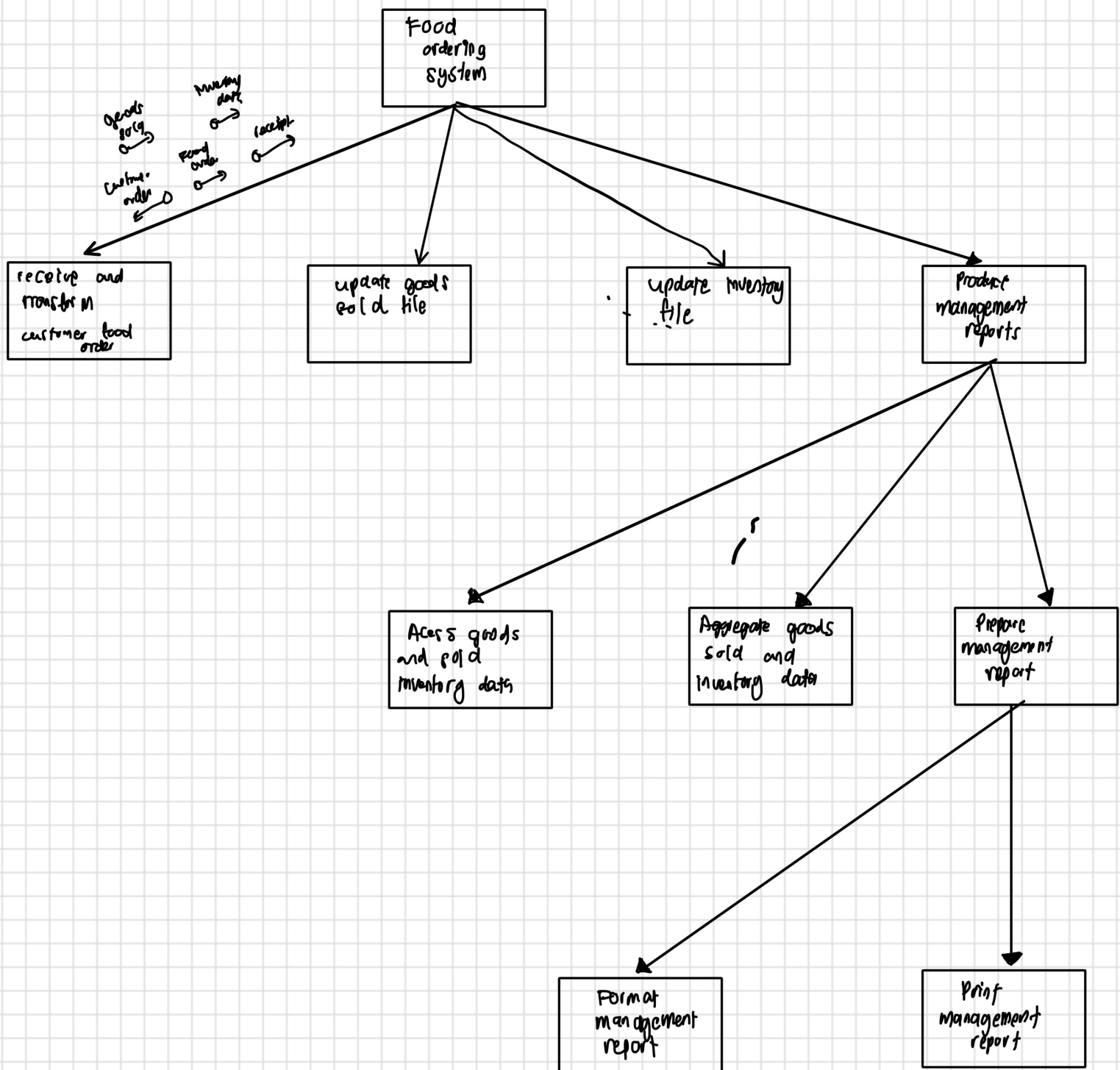
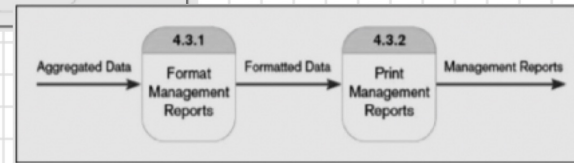
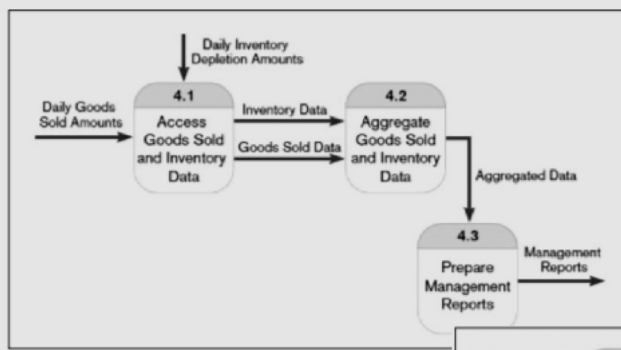
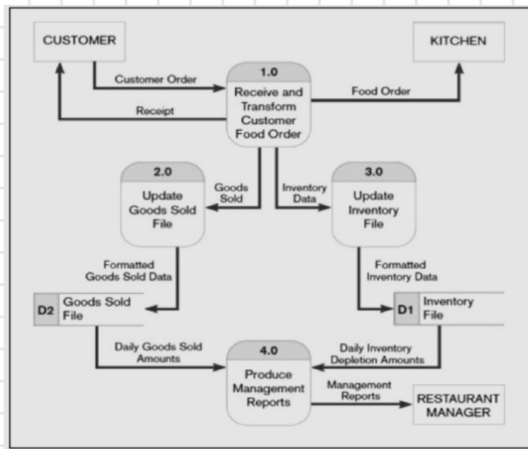


Figure 1(b): Level 1 Diagram for PROCESS 4.0

- a) - They both use different symbols to represent component
 - DFD is to show data movement while structured chart shows the structural hierarchy of modules

b)



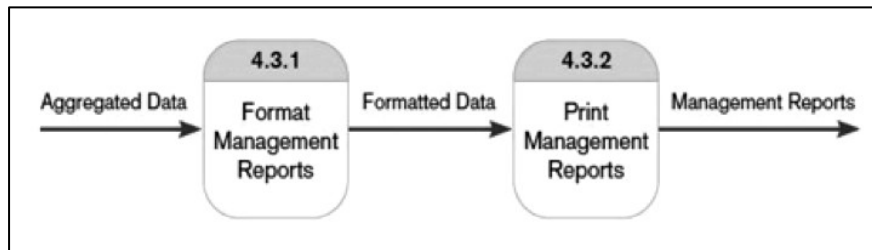


Figure 1(c): Level 2 Diagram for PROCESS 4.3

QUESTION 3 (20 marks)

Speedy Spuds is a fast-food restaurant offering all kinds of potatoes. The manager has a 30-second rule for serving customers. They have an ordering system (Speedy Spuds Order System), which was developed 20 years ago when they started the business.

The system is used by the *Servers* (staff who took order from customers), the *Kitchen Crews* (staff who will prepare the order) and the *Waiters* (staff who serve the order). At the counters, orders are taken by Servers by filling in a service request form (Figure 2):

SPEEDY SPUDS SYSTEM

12 December 2019 12:12:12PM

SUBMIT ORDER

FILL IN ORDER. PLEASE TYPE ORDER

CUSTOMER NAME: _____

SEARCH CUSTOMER

CUSTOMER NOT FOUND? **NEW CUSTOMER**

POTATO (choose from *BAKED/FRIED/MASHED/ROASTED*): _____

TOPPINGS (choose from *CHEESE/SALT/PAPRIKA/TOMATO SAUCE*): _____

SIZE: SMALL (RM4.00) / MEDIUM (RM7.00) / LARGE (RM12.00)

QUANTITY: _____

TOTAL PRICE: _____

(For topping Price add RM2 each except Tomato Sauce):

Figure 2: Speedy Spud Order System

The current form is difficult for Servers to fill out quickly. The servers say they could achieve the 30-seconds rule if the form were simplified.

- a) Design a form that lists possible potatoes and toppings in a manner that is easy for counter servers to fill in. Justify all the guidelines for your design.
(Hint: remember to apply all the guidelines for good form design.)

(15 marks)

- b) Speedy Spuds management team are proposing to upgrade the Speedy Spuds System to include a monthly report for **inventory and reorder supply system**.
- i. Printed report design falls into **THREE (3)** categories. List ALL of the report design categories.

(3 marks)

- ii. Which of the report design in (i) that you think is suitable for Speedy Spuds' monthly report for inventory and reorder supply system? Justify your answer.

(2 marks)

Question 3

a)

Customer name: [text box] [button]

order details

Potato type:	Toppings:	Sauce:
☐ Bacon	☐ Cheese	☐ ..
☐ Fries	☐ ..	☐ ..
☐ Mashed	☐ ..	☐ ..
☐ Frost	☐ ..	☐ ..

Quantity: [text box] [button] Submit [button] Total Price [text box]

1. the order detail are designed with proper flow, from left to right and top to bottom.
2. The information are grouped logically with proper placement of heading, body and others.
3. The form are easy to fill out, server only need to click on the applied checkbox.
4. Clear caption and instruction are provide to ensure accurate completion.
5. section where the users need
6. the "submit order" button are placed clearly at the bottom of the form.

b)

- i) - detailed report
 - Exception report
 - Summary report

- ii) - Exception report
 - Suitable for quick to-read report for busy restaurant but still provide sufficient details.

QUESTION 4 (15 marks)

An organization called Sparkling Energy Positioning and Knowledge (SEPAK) is planning to install an enhanced and up to date Information Systems to replace their current systems in every branch they have in Malaysia. The project has been outsourced to your company. Now, the system development already completed, and you are planning for the system installation. However, their IT manager, Mr. Ragaman, requires the installation process to be low risk and it is the only matter he concerned because the system stores very important data and run crucial processes for the organization.

- a) Propose **THREE (3) installation strategies** that is suitable for the above situation. Elaborate your answers.

(9 marks)

- b) After reviewing all the proposed installation strategies, Mr. Ragaman also requires that the installation process to allow the IT Department to compare the performance of the new system with the old one.

Which strategy from (a) you think is the most suitable? Justify your answer.

(4 marks)

- c) Preparing user support by providing ongoing educational and problem-solving assistance is one of the important parts in the implementation process. One of the current approaches is by using automation method, where it can reduce cost on user support. State **TWO (2) examples of automation methods** in user support.

(2 marks)

a) Parallel installation

- ensure the system can run properly before full installation
- ensure important data saved in the old system and slowly transferred to new system
- crucial process can still run if the new system has problem.

2) Single location installation

- low risk for different location
- bad effect or any issue won't affect whole organization

3) Phased installation

- can try for less important module first to assess risk
- bad effect or issue wouldn't affect whole system and can be solved more easily
- low risk

b) parallel installation

- can compare result of new system with the old system
- can see any obvious difference or issue compared to old system effectively
- can compare which system has better performance
- useful for comparison until the old system are decided to be turned off

c) - voices response system

- knowledge bases

PART B : CASE STUDY (30 MARKS)

INSTRUCTION : ANSWER ALL QUESTIONS.

Please write your answer in the answer sheets.

All Phones Shop (APS) is a company that sells different types of phones, including smartphone, mobile phone, and land phone. The company developed a system that helps them keep track of their clients. The system includes these processes:

- To buy or sell a phone, the client must first register. In order to register, the client has to provide their IC number, name and address. However, APS will check first if the client already exists based on their IC num. If the IC number is not in the system, it will create a unique registration ID for the client and add a record in the client file.
- APS helps the clients to buy and sell any phone that they want. To sell an item (phone), the client first selects the type of the item that he/she wants to sell. Then the client provides model, make, capacity, and colour of the phone. APS will assign an ID to the phone, records and attaches the phone information with the client registration ID. This will be store also in the product file.
- For each of the phone to be sell, APS will create a record in the catalogue file. Here, the APS system will automatically create a display image based on the description, asks the client (seller) to approve it. Once the client approves the display image, APS includes it in its product catalogue file.
- Clients can also buy phones as recorded in the product catalogue file. The client will create a shopping basket for the selected phone(s). Then after the client checks out, and each phone in the basket is considered “sold” and removed from the product catalogue file.
- APS prepares an invoice for the sold item. Once the client (buyer) confirms the invoice, it is attached with the client registration with the status “unpaid”. The payment can be made either using online transfer or credit card. After the payment is confirmed, the invoice status is changed to “paid” and will notify the seller.

The Context Diagram for APS is as follows:

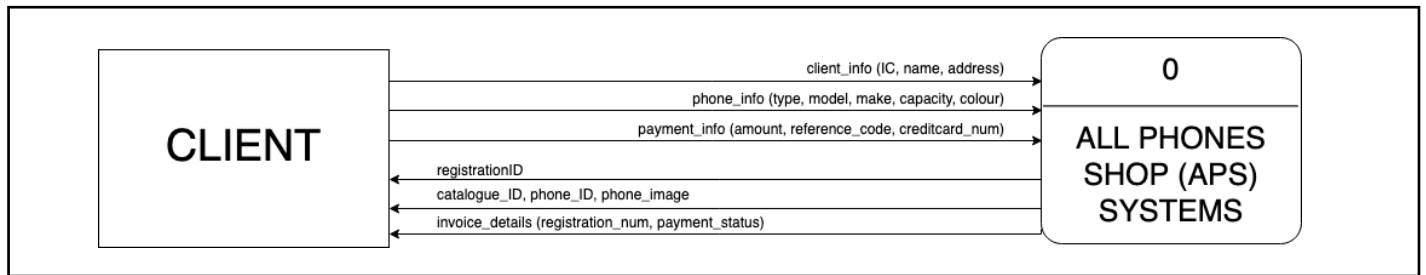


Figure 3: APS System Context Diagram

Based on the case study and Figure 3, answer all the questions below:

- Draw the Level 0 Diagram based on context diagram in Figure 3.
(10 marks)
- One of the process in (a) is “**SELL A PHONE**”. Draw Level 1 PHYSICAL DFD of the process.
(10 marks)
- Partition the Level 1 Physical DFD in (b) by separating and grouping the appropriate processes. You can answer this in the same diagram as (b).
(4 marks)
- Justify your answer in (c). Explain why you choose the processes to be partitioned.
(6 marks)